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University of Virginia

When Do Explanations Explain? A Controlled Study of XAI Under Varying Signal-to-Noise

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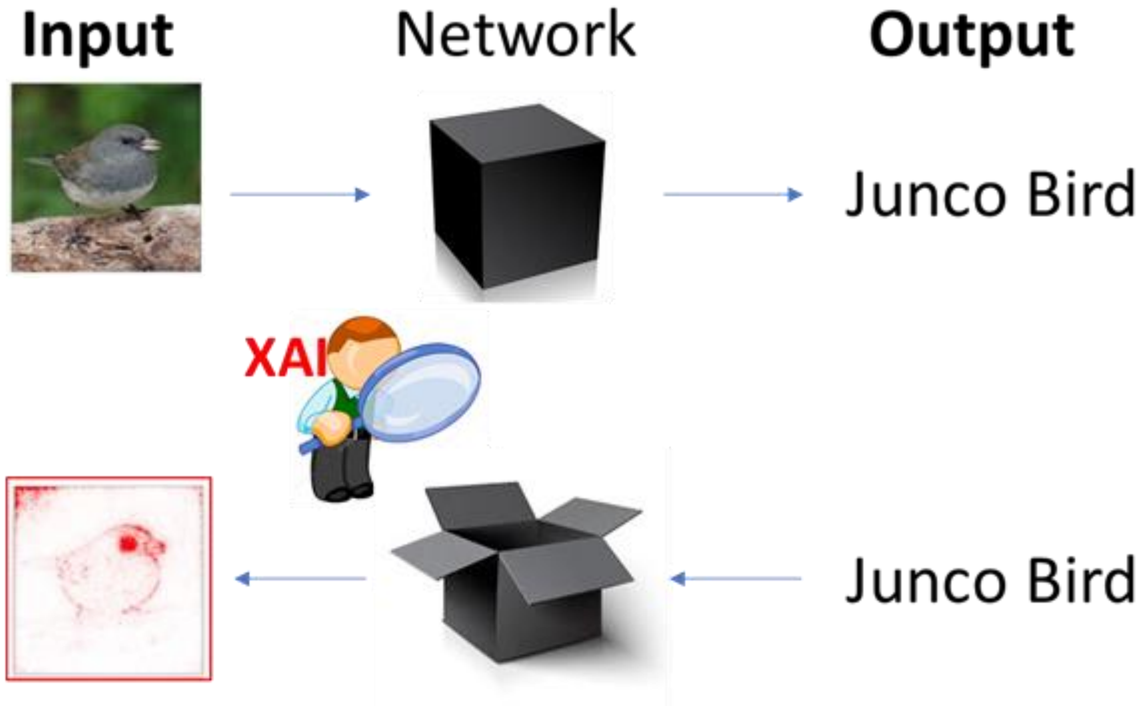
Brief Introduction to Explainable Artificial Intelligence (XAI)

What is XAI and why do we need it?

Why is XAI necessary?

Methods of **eXplainable Artificial Intelligence (XAI)** aim to explain how a Neural Network makes predictions, i.e., what the *decision strategy* is.

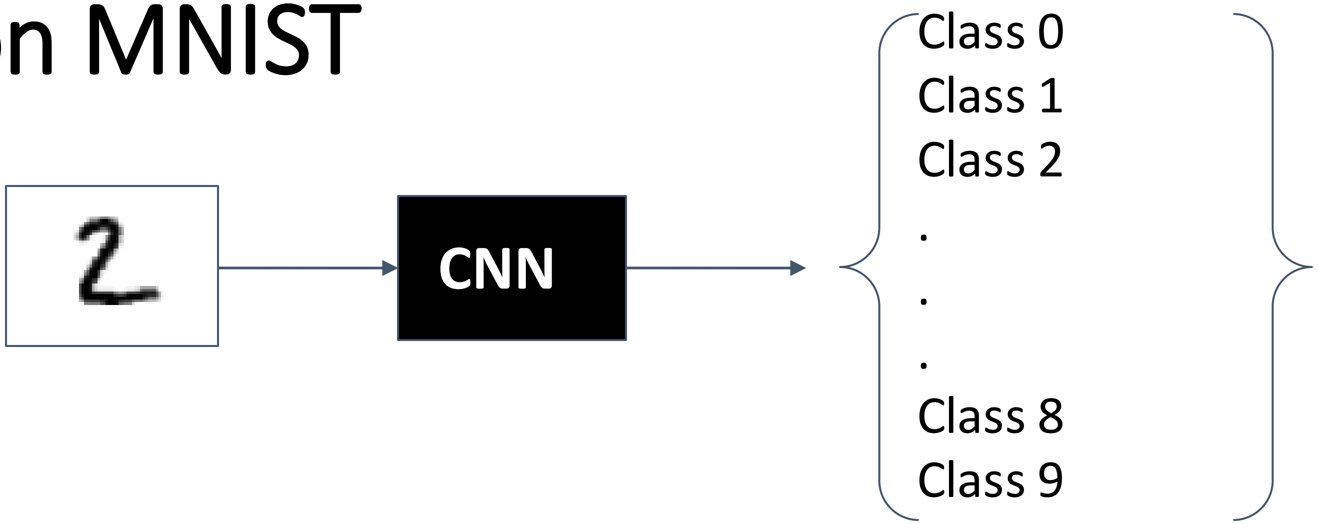
XAI methods highlight which features in the input space are important for the prediction: They produce the so-called *explanation/relevance heatmaps.*

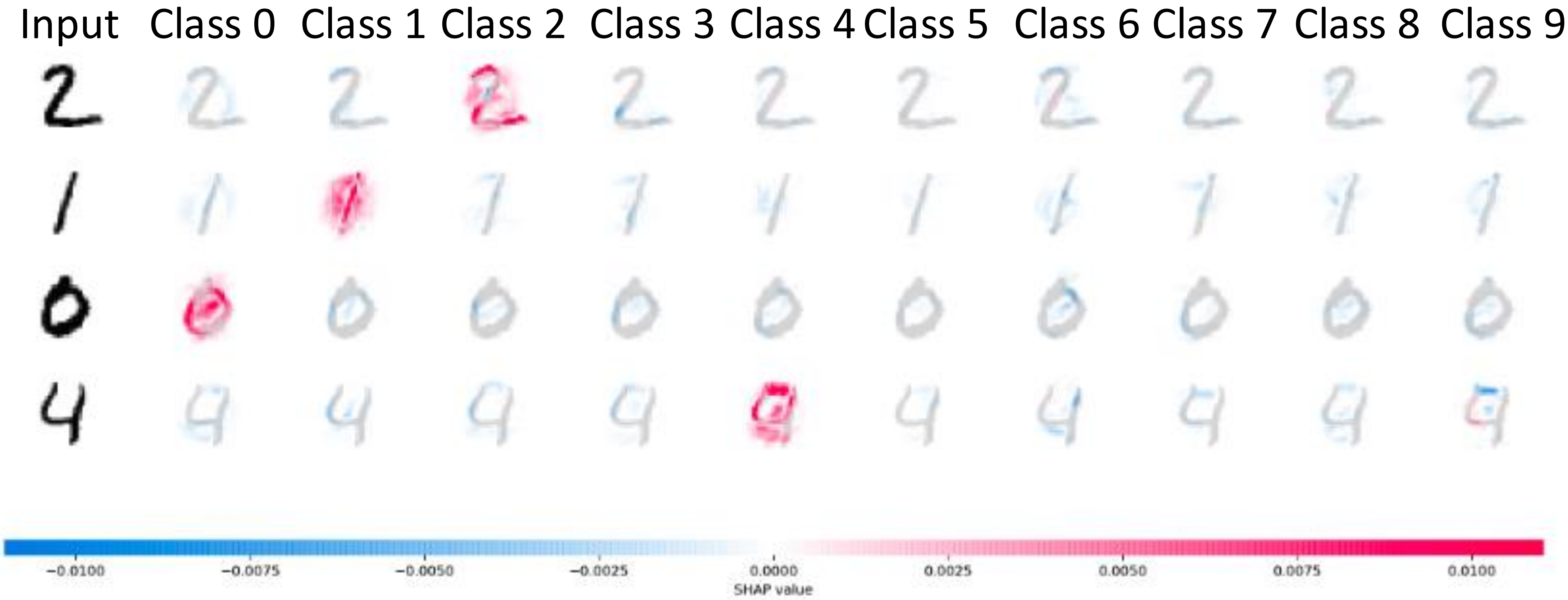


Explanation (or relevance) Heatmap

From *Adebayo et al. (2020)*

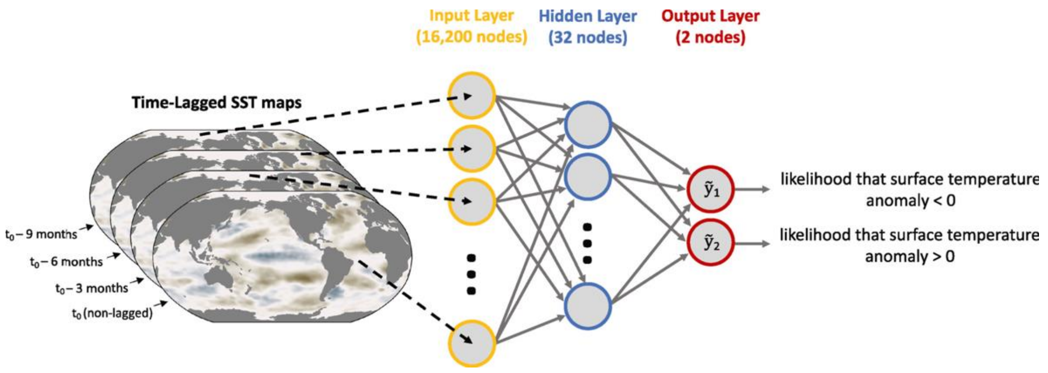
XAI Example on MNIST

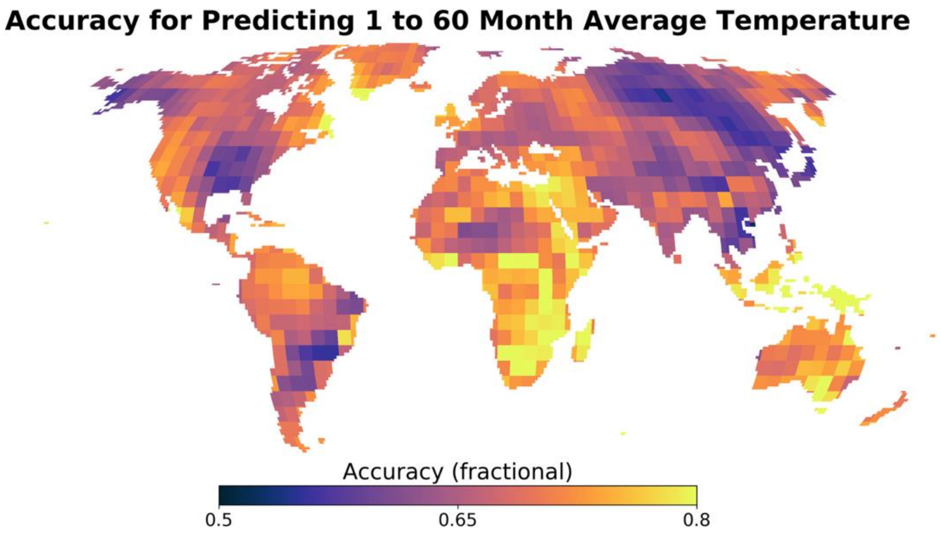




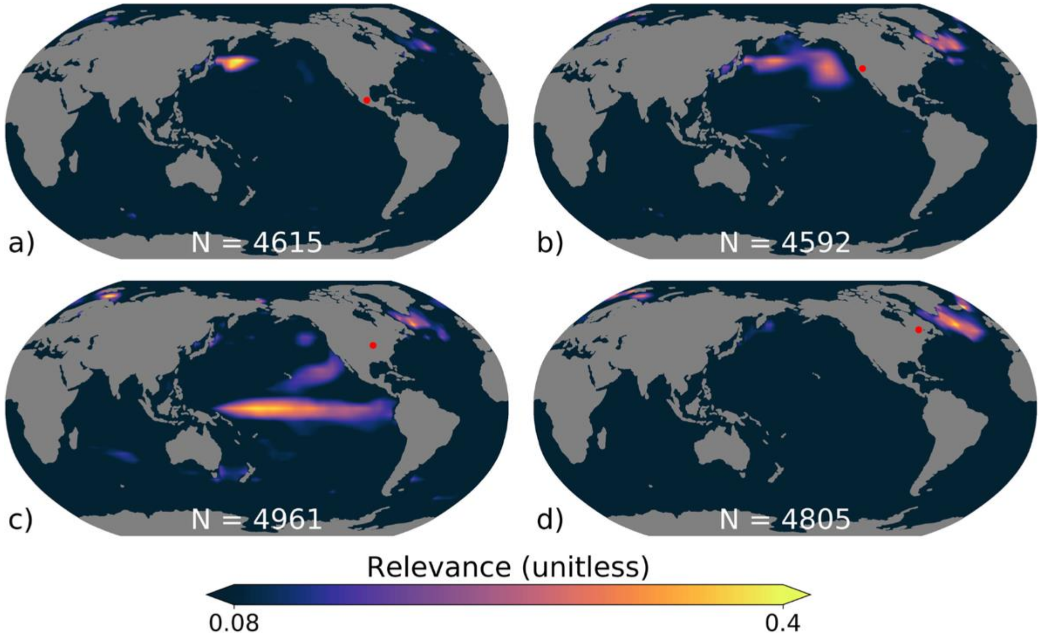
XAI Example on Decadal Predictability in the Earth System

Monthly anomalies of Sea Surface Temperatures are used to predict the sign of SST anomaly over land in the next 5 years.





XAI results:



From *Toms et al. (2021)*

XAI Benchmarks

How to objectively assess XAI methods?

The need for *objectivity* in assessing XAI

Which input features were important for this classification?

XAI method

XAI heatmap

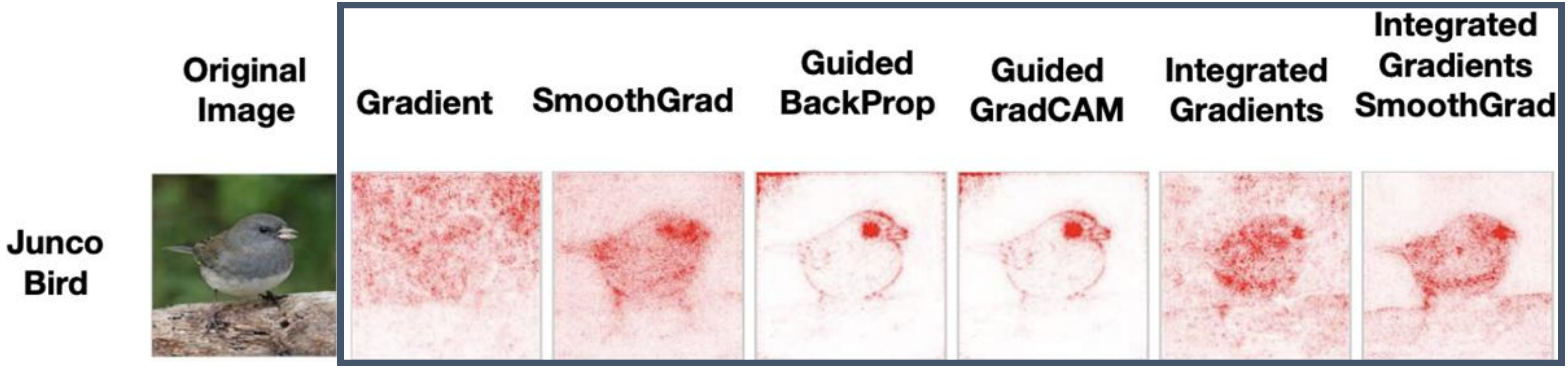
Issues : 1) No ground truth to assess the estimated explanations.

From *Adebayo et al. (2020)*

The need for *objectivity* in assessing XAI

Which input features were important for this classification?

Many Different XAI methods

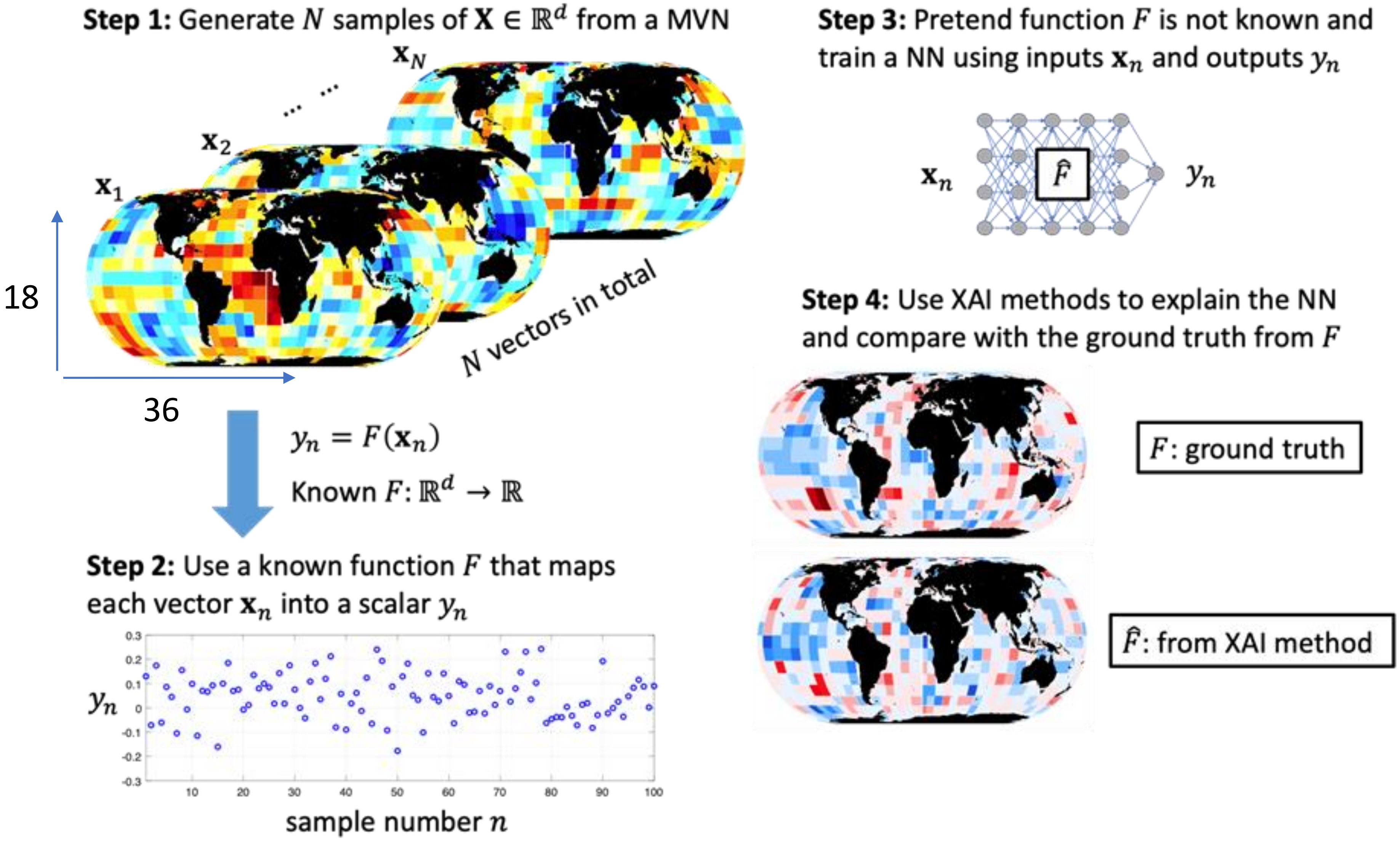


Issues : 1) No ground truth to assess the estimated explanations.
2) Different methods provide different answers.

- **This is problematic: The uncertainty on how the network decides, leads to limited trust when using neural networks in environmental problems.**
- **We need objective frameworks to rigorously assess XAI methods and gain insights about relative strengths and weaknesses.**

From *Adebayo et al. (2020)*

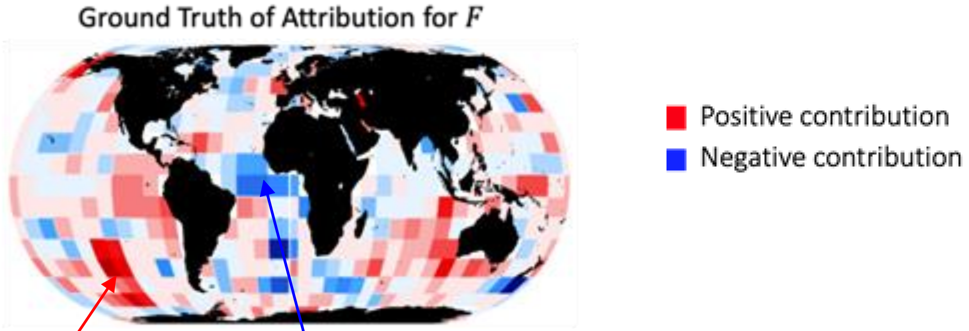
Attribution benchmarks for XAI



From *Mamalakis et al. (2022a)*

Attribution benchmarks for XAI

y_n : 0.0660
NN prediction: 0.0802



Red color highlights features that contributed positively to y

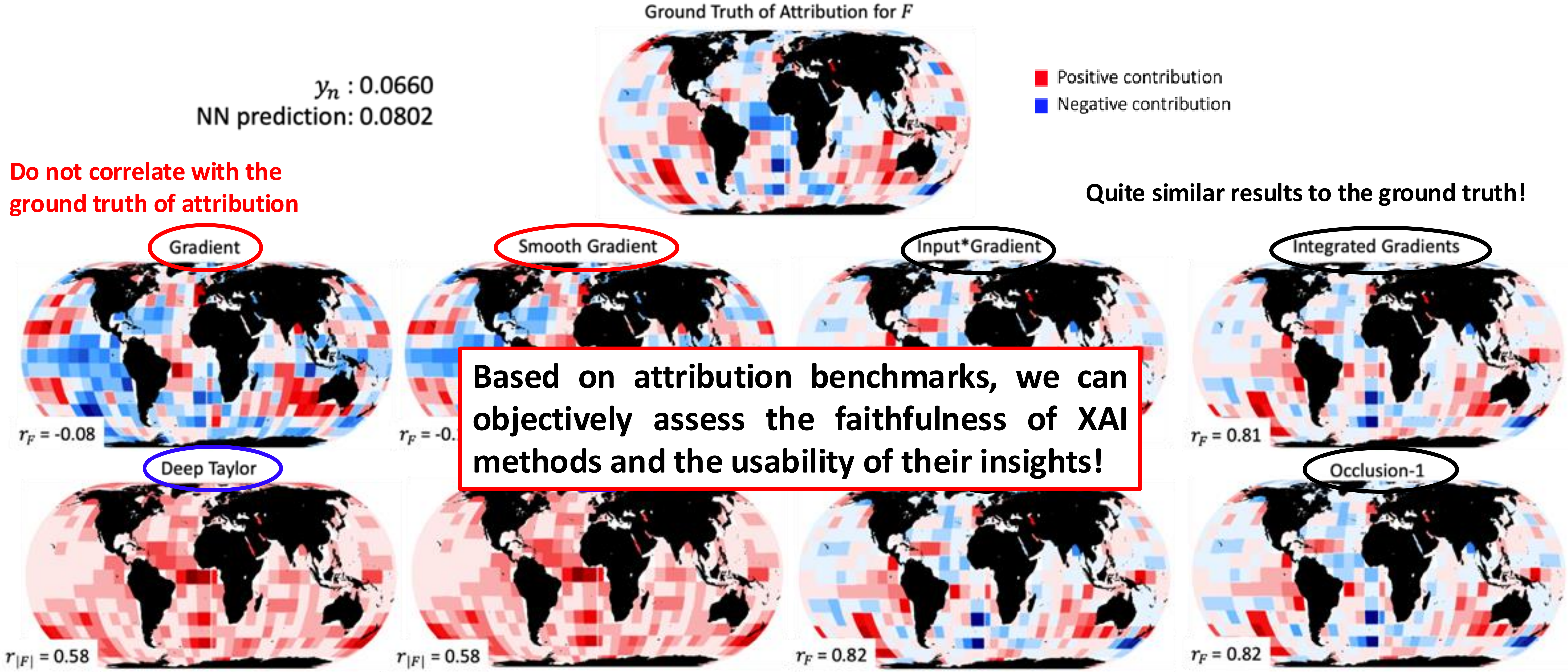
Blue color highlights features that contributed negatively to y

From *Mamalakis et al. (2022a)*

Attribution benchmarks for XAI

Do not correlate with the ground truth of attribution

Quite similar results to the ground truth!



Based on attribution benchmarks, we can objectively assess the faithfulness of XAI methods and the usability of their insights!

Provide only positive attributions. Cannot distinguish the true sign of attribution.

From *Mamalakis et al. (2022a)*

Realistic conditions:
Noise in the data, limited sample size

Having built a controlled synthetic benchmark, we can next ask:

How do different degrees of noise in the data and limited data sizes affect the fidelity and usability of XAI insights?

Mamalakis, Antonios, William E Chapman, Kirsten J Mayer. Expanding the Limits of Explainable AI Tools in Generating Physical Insights. *TechRxiv.* November 03, 2025. **DOI:** 10.36227/techrxiv.176221224.44249956/v1

Three levels of noise

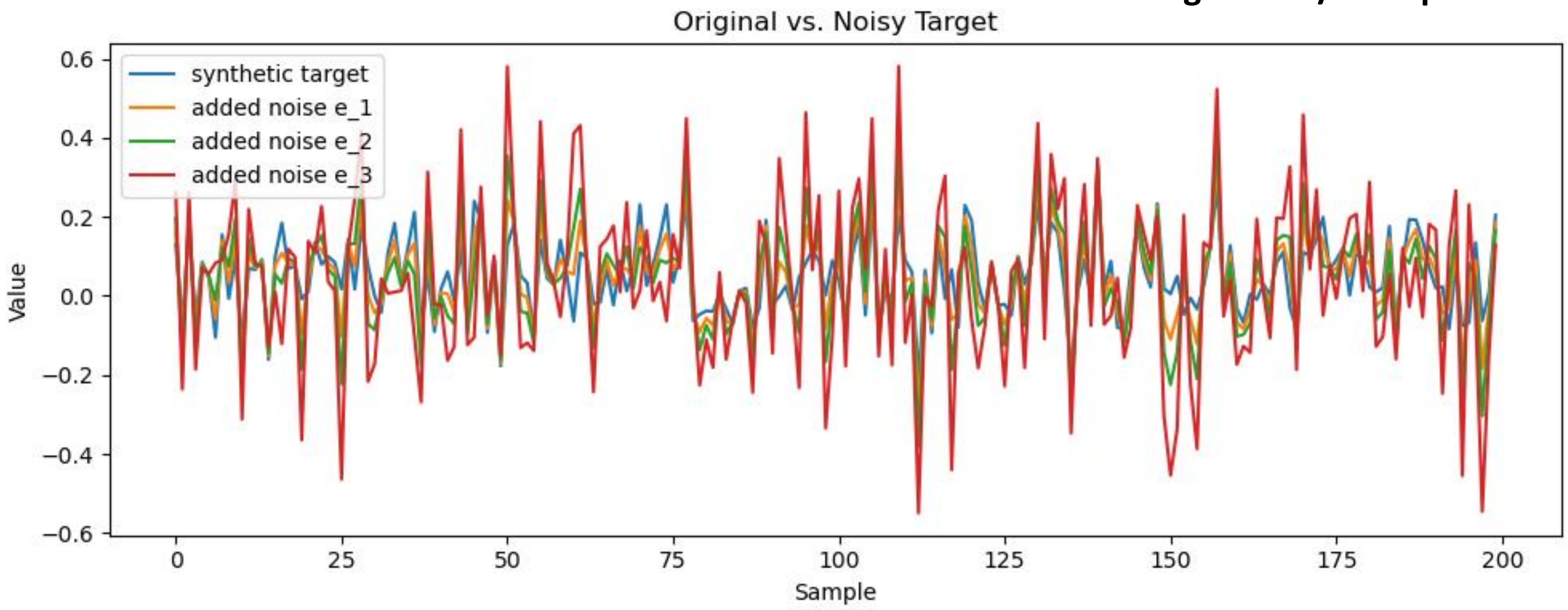
From *Mamalakis et al. (2025)*

0) Noise free: y (unrealistic) → **Noise-free/Data-rich Model**

1) Adding **low noise**: $z_1 = y + e_1$, with $\boldsymbol{e_1} \sim$ **N (0, 0.5σ$_y$)**
   - **Low-noise/Data-rich Model:** Train with 100,000 samples
   - **Low-noise/Data-poor Model:** Train with 1,000 samples

2) Adding **moderate noise**: $z_2 = y + e_2$, with $\boldsymbol{e_2} \sim$ **N (0, σ$_y$)**
   - **Moderate-noise/Data-rich Model**
   - **Moderate-noise/Data-poor Model**

3) Adding **high noise**: $z_3 = y + e_3$, with $\boldsymbol{e_3} \sim$ **N (0, 2σ$_y$)**
   - **High-noise/Data-rich Model**
   - **High-noise/Data-poor Model**



Model Performance Across Experiments

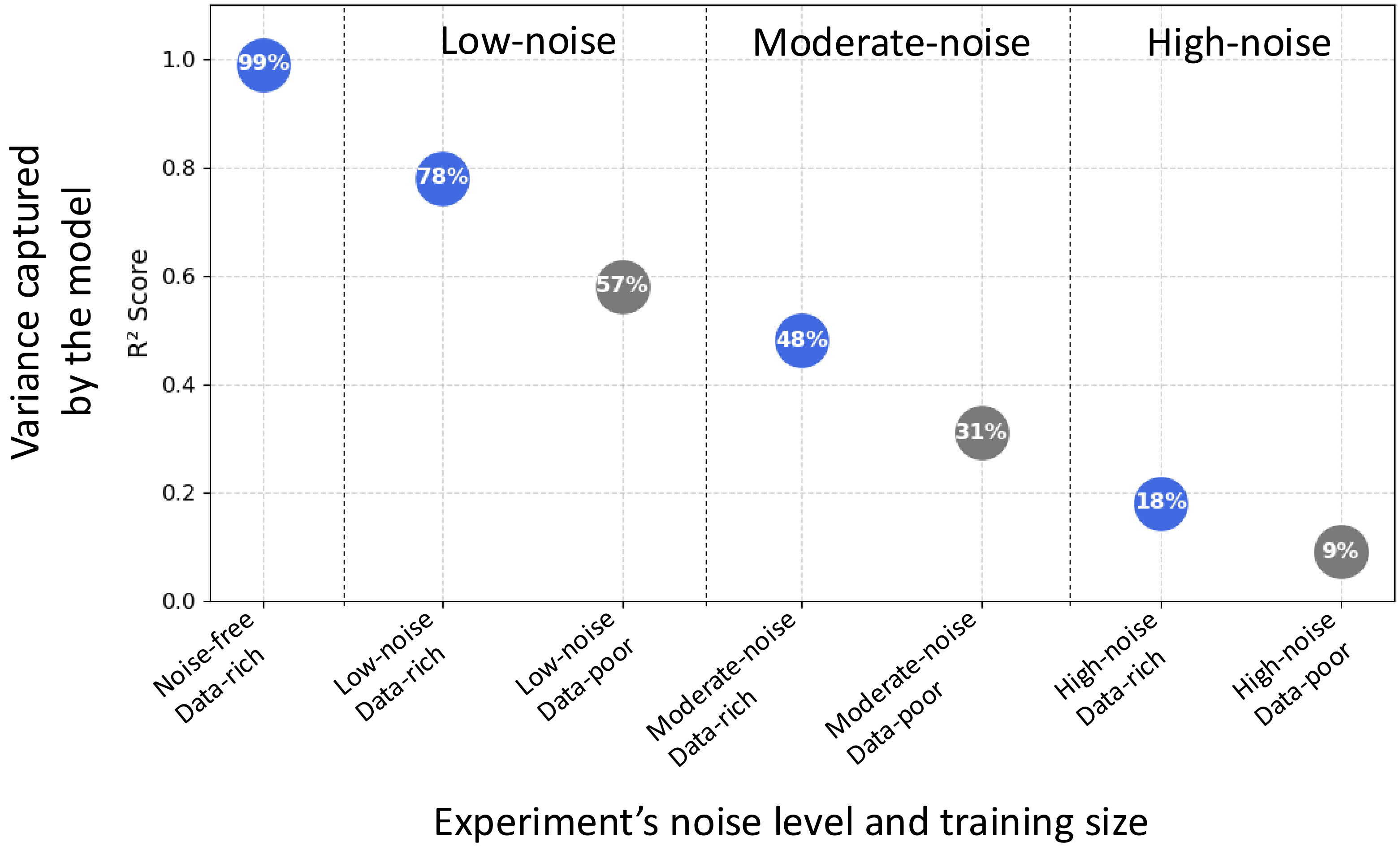
From *Mamalakis et al. (2025)*

Using 100,000 training samples **(Data-rich)**

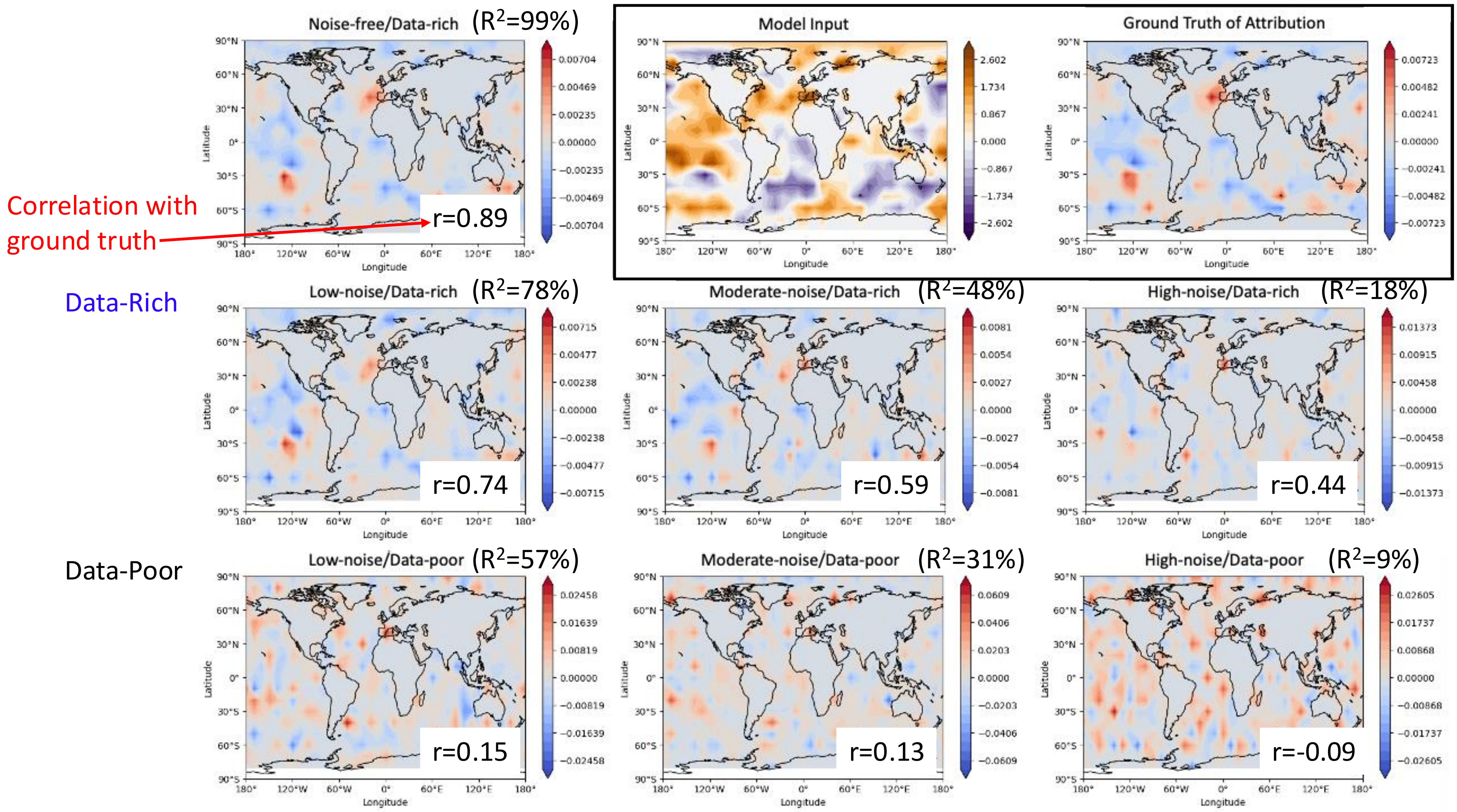
Using 1,000 training samples **(Data-poor)**

✓ Higher noise leads to lower R^2.

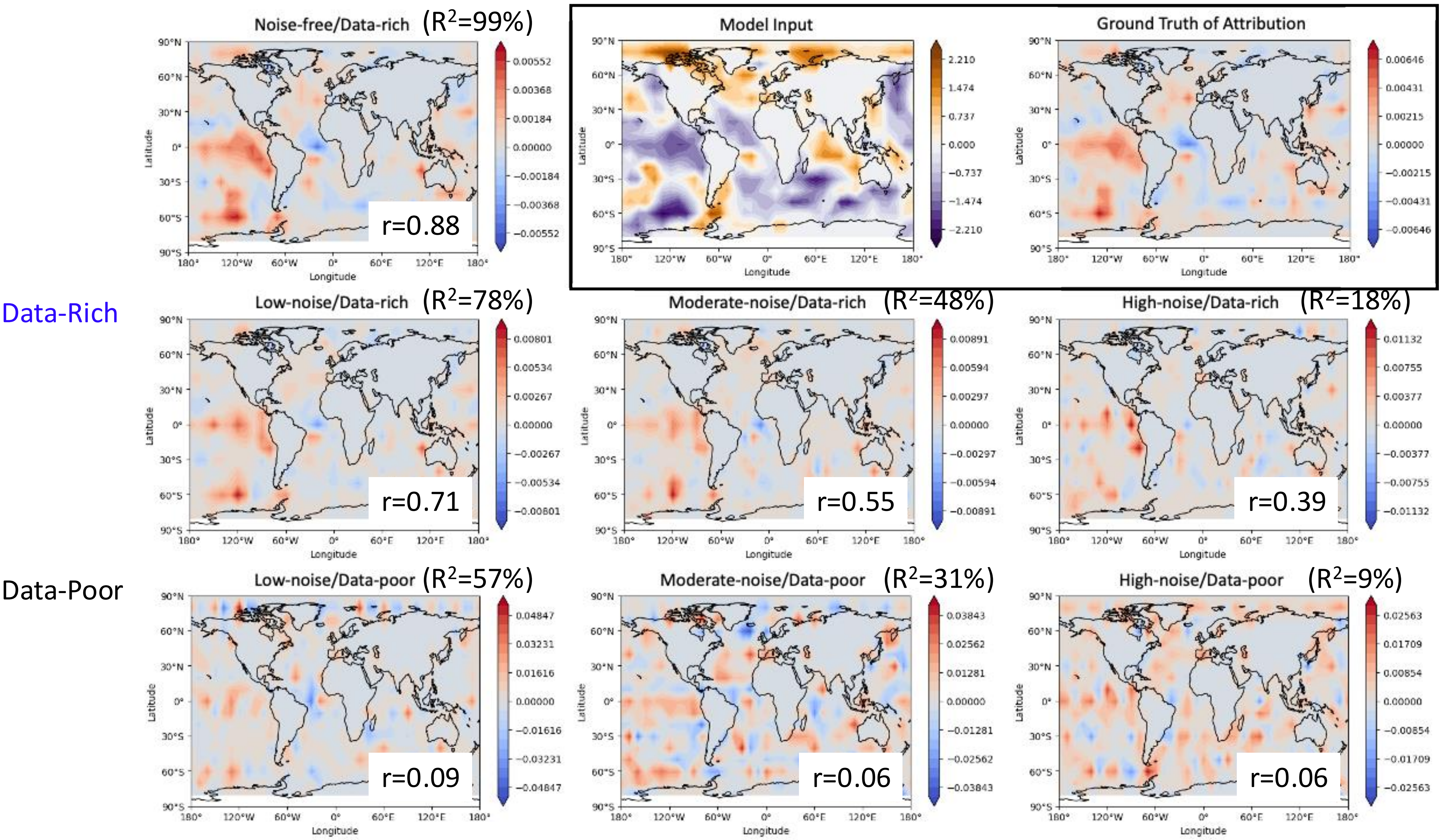
✓ Given a noise level, more data leads to higher R^2.



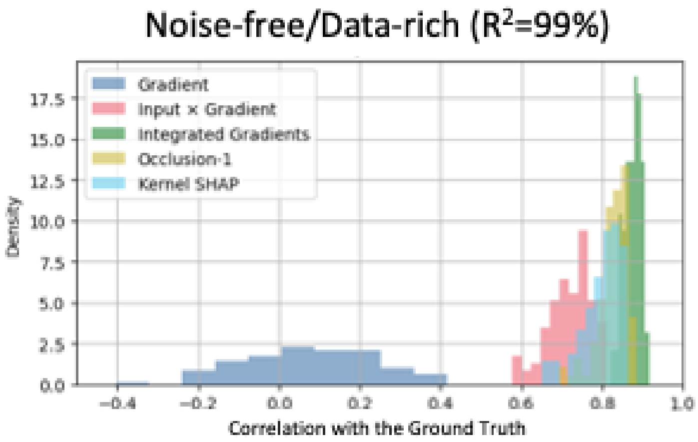
Integrated Gradient Heatmaps for a second test sample



Integrated Gradient Heatmaps for a representative test sample



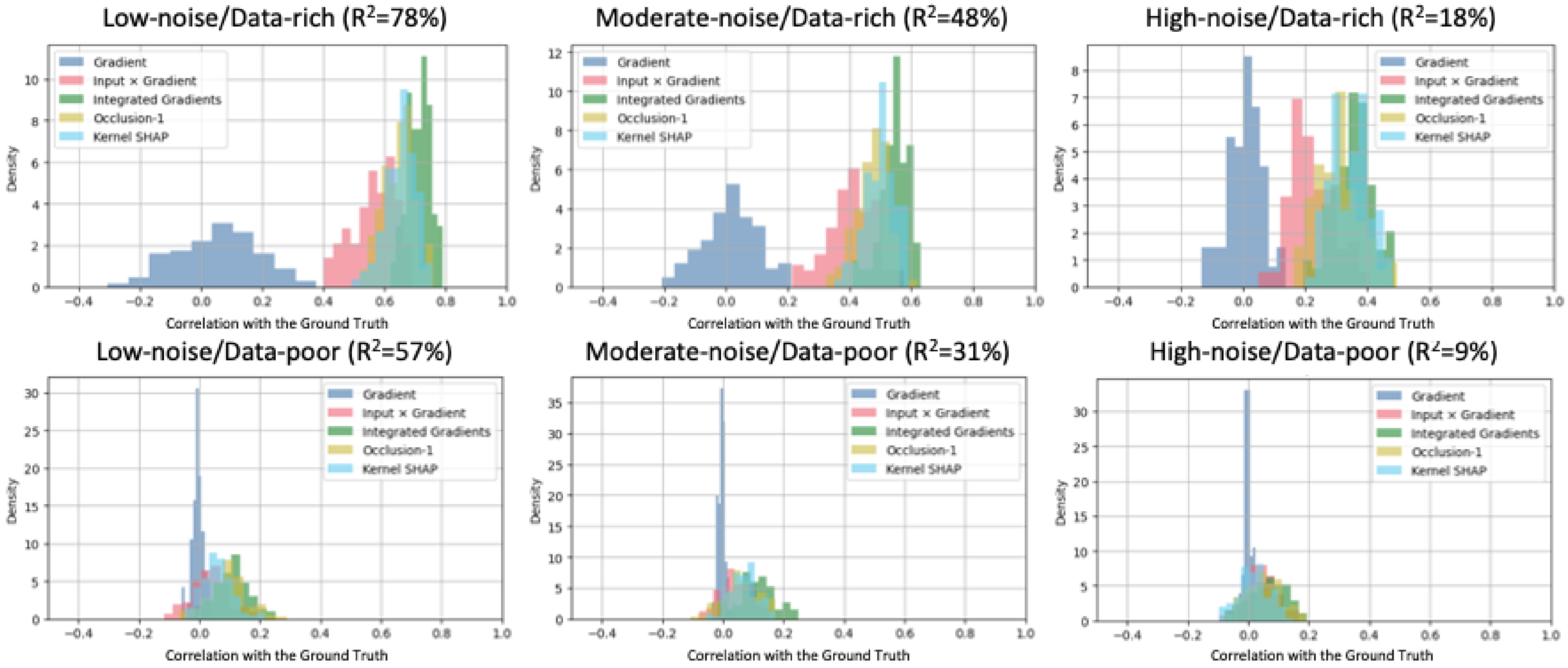
Probability Distributions of Correlation with the Ground Truth



To gain valuable XAI insights the raw model performance (R^2) is irrelevant!

Data-Rich

Data-Poor



Learnable Variance

From *Mamalakis et al. (2025)*

Let's consider the low-noise experiment, with $z_1 = y + e_1$ and $e_1 \sim N\left(0{,}0.5 * \sigma_y\right)$ is the added noise. Then the variance of z_1 is:

$$Var(z_1) = Var(y) + Var(e_1)$$

, as y and e_1 are uncorrelated

$$= Var(y) + \frac{1}{4}Var(y)$$

$$= \frac{5}{4}Var(y)$$

And the learnable faction of the variance is:

$$\frac{Var(y)}{Var(z_1)} = \frac{Var(y)}{\frac{5}{4}Var(y)} = \mathbf{80\%}$$

So, for the **low-noise experiment**, **80% of the variance in the target is learnable** (deterministic signal) and **20% is unlearnable** (noise)!

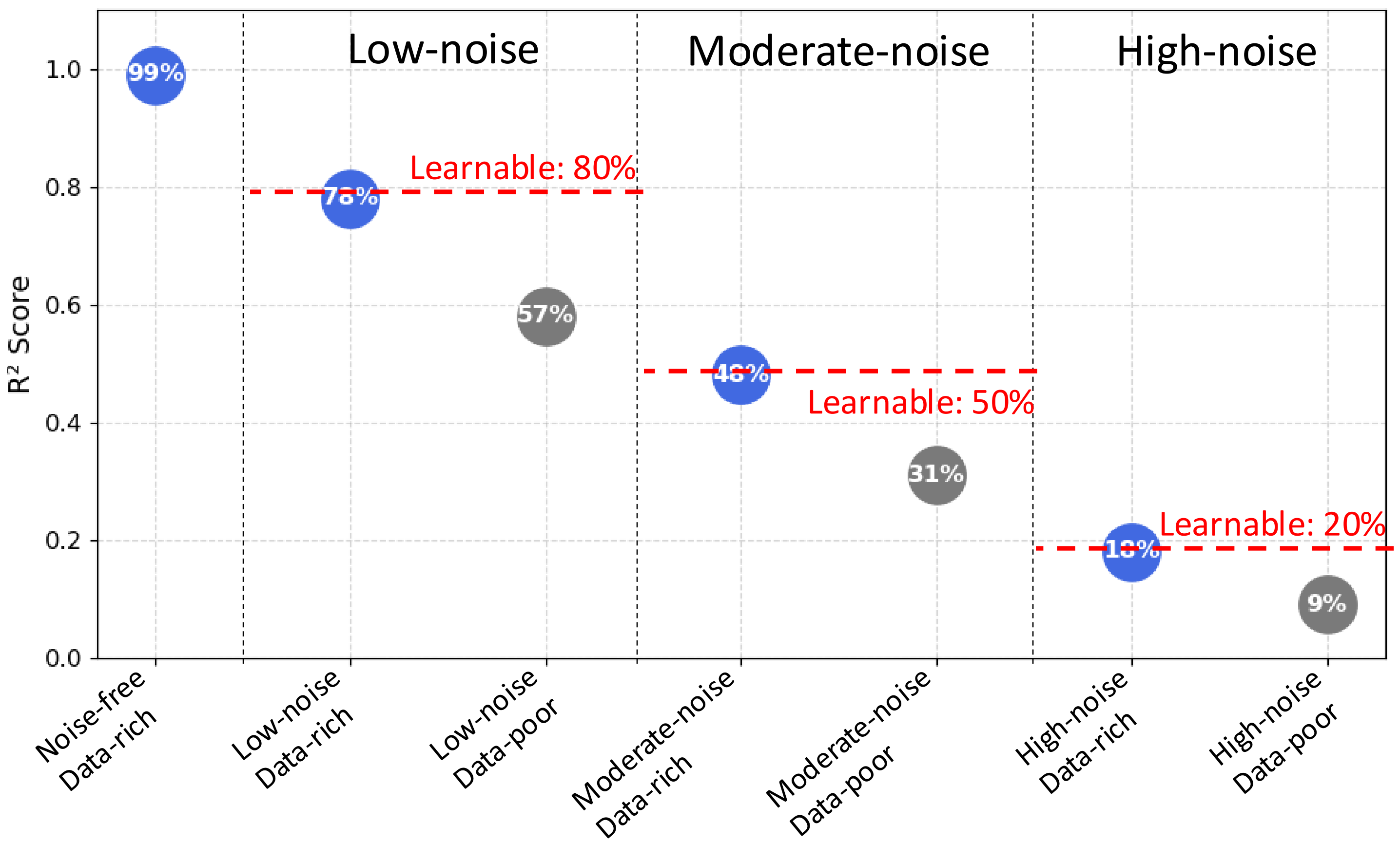
Model Performance Across Experiments

From *Mamalakis et al. (2025)*

Using 100,000 training samples **(Data-rich)**

Using 1,000 training samples **(Data-poor)**

Variance captured by the model



Experiment's noise level and training size

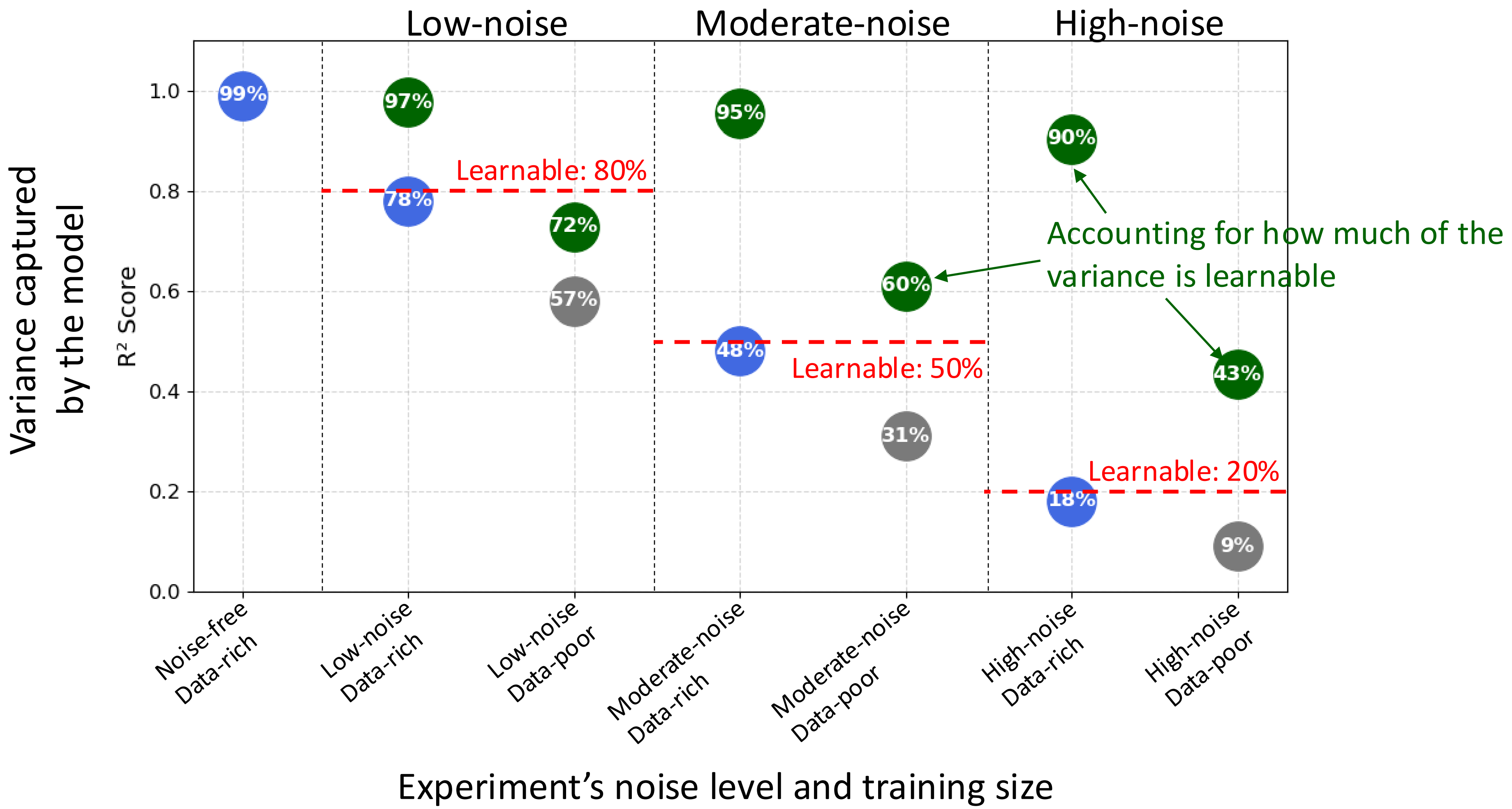
Model Performance Across Experiments

From *Mamalakis et al. (2025)*

Using 100,000 training samples **(Data-rich)**

Using 1,000 training samples **(Data-poor)**

Rescaled R^2 to account for the learnable variance



The need for proxies beyond prediction performance

- High (or low) R^2 does not imply high (or low) XAI fidelity and reliability. What matters is how much of the ***learnable signal*** the model has captured!

- In practice, we never know how much of the learnable signal our model has captured or how much training size is sufficient because these depend on the complexity of the task and the level of noise in the data.

  So, how will we know if our AI models have learned anything?

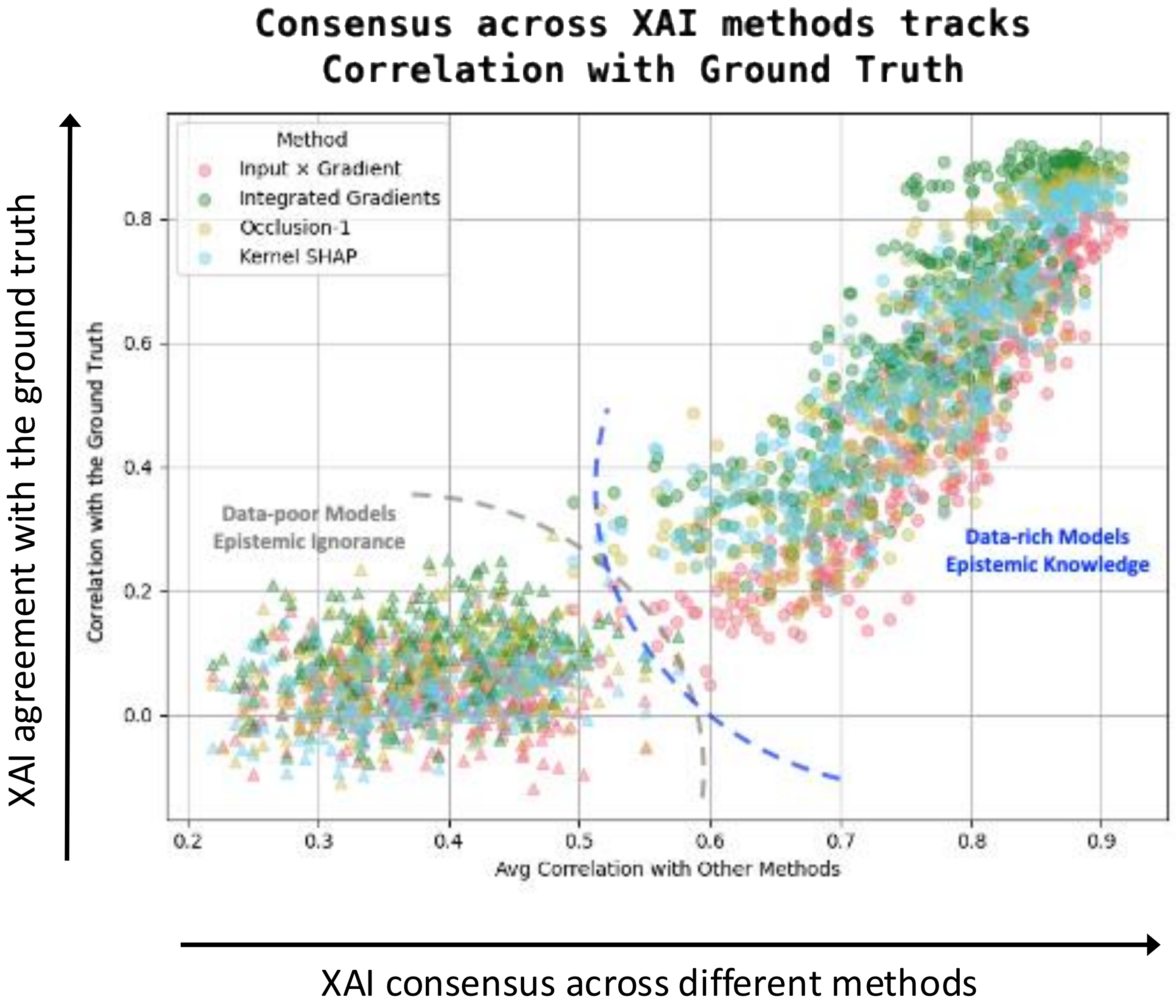
From *Mamalakis et al. (2025)*

Useful Proxies – Inter-method XAI Consensus

Two distinct clusters form!

1) **Data-rich models:** Inter-method XAI consensus tracks the agreement with ground truth.
These models have entered a regime of **epistemic knowledge** and show evidence of **meta-cognition**!

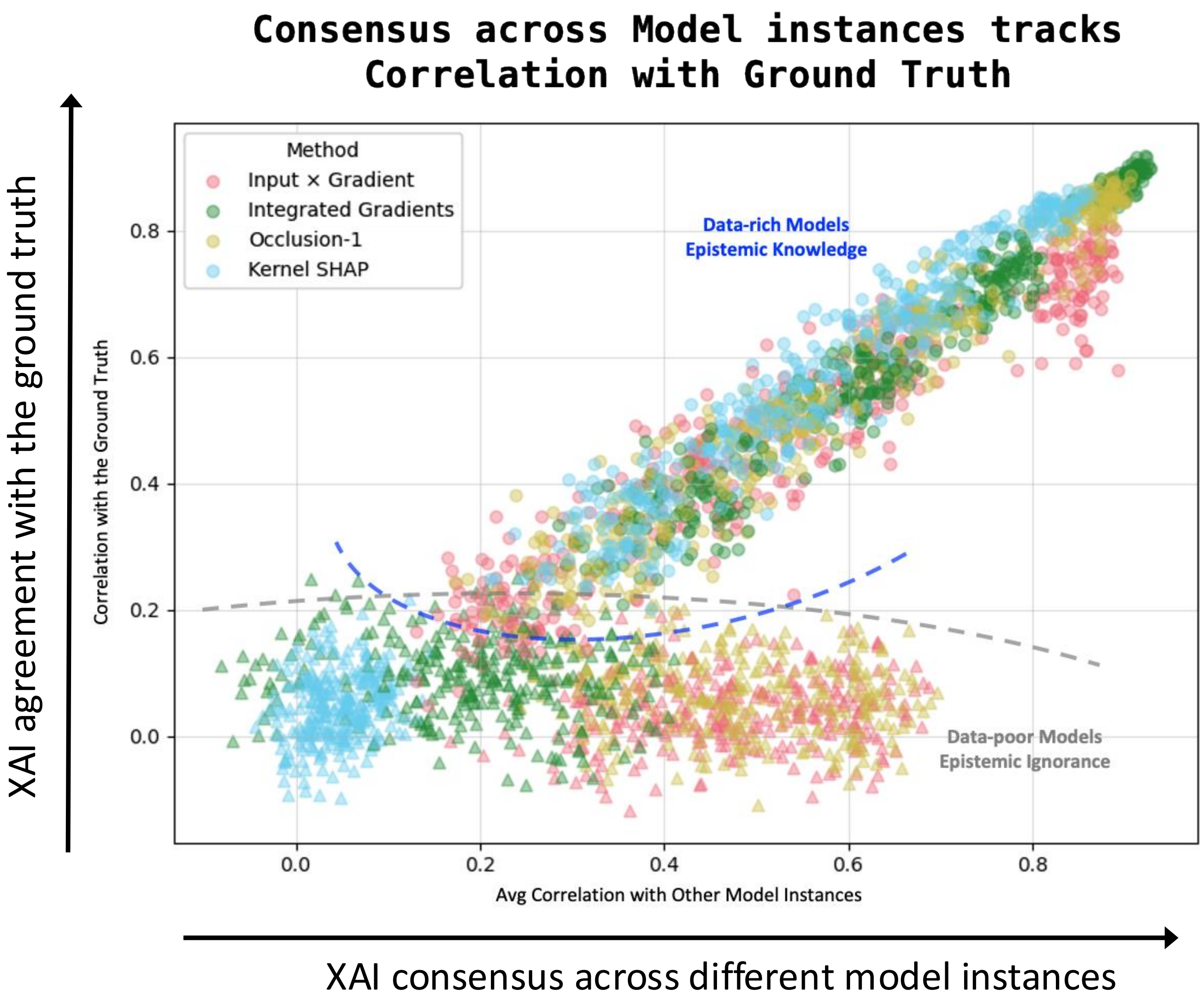
2)**Data-poor models:** Inter-method XAI consensus is irrelevant to the agreement with the ground truth!
These models remain in a state of **epistemic ignorance!**



From *Mamalakis et al. (2025)*

Useful Proxies – Inter-model XAI Consensus

Again here, two distinct clusters form!
Similar insights as before apply!

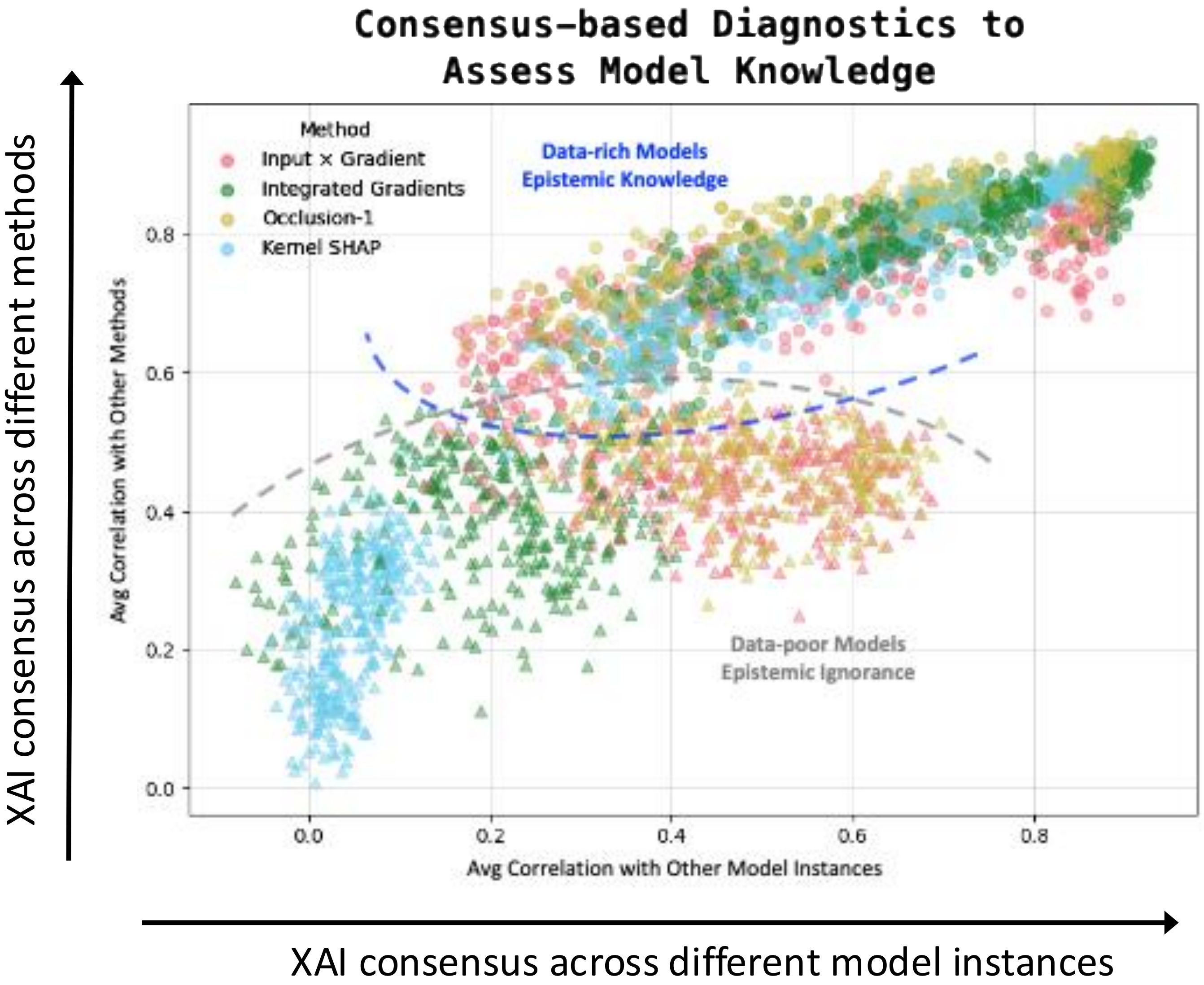


From *Mamalakis et al. (2025)*

Linear structure of consensus in knowledgeable models

In knowledgeable models, there is a **linear dependence between the inter-model and inter-method consensus** across samples!

This is a meta-diagnostic (it's the correlation of the correlations) to assess XAI fidelity in real applications!



From *Mamalakis et al. (2025)*

Take-home Messages

- **Attribution benchmarks** are among the very few ways we can objectively assess the faithfulness of XAI methods and the usability of their insights across different training settings and tasks (realistic or not).

- High (or low) R^2 does not imply high (or low) XAI fidelity and reliability. What matters is how much of the ***learnable signal*** the model has captured!

- Our results point to the **generation of XAI ensembles** as a best-practice approach to assess XAI usability. Generate many XAI heatmaps (from different XAI methods or models). Then, **explore XAI consensus**: XAI consensus is likely to exhibit a linear structure in epistemically knowledgeable models, while such structure is unlikely in ignorant models.

"

If you ever wonder whether your AI has truly learned something, don't look at its accuracy.
Build an XAI ensemble, and let the consensus speak.

Mamalakis, Antonios, William E Chapman, Kirsten J Mayer. Expanding the Limits of Explainable AI Tools in Generating Physical Insights. *TechRxiv.* November 03, 2025. **DOI:** 10.36227/techrxiv.176221224.44249956/v1

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Thank you!

References

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Mamalakis, A., I. Ebert-Uphoff and E.A. Barnes (2022a) “Neural Network Attribution Methods for Problems in Geoscience: A Novel Synthetic Benchmark Dataset”, arXiv preprint, https://arxiv.org/abs/2103.10005, *Environmental Data Science.*

Mamalakis, A., E. A. Barnes, and I. Ebert-Uphoff, 2022b: Investigating the fidelity of explainable artificial intelligence methods for applications of convolutional neural networks in geoscience. arXiv, 2202.03407v2, https://doi.org/10.1175/AIES-D-22-0012.1.

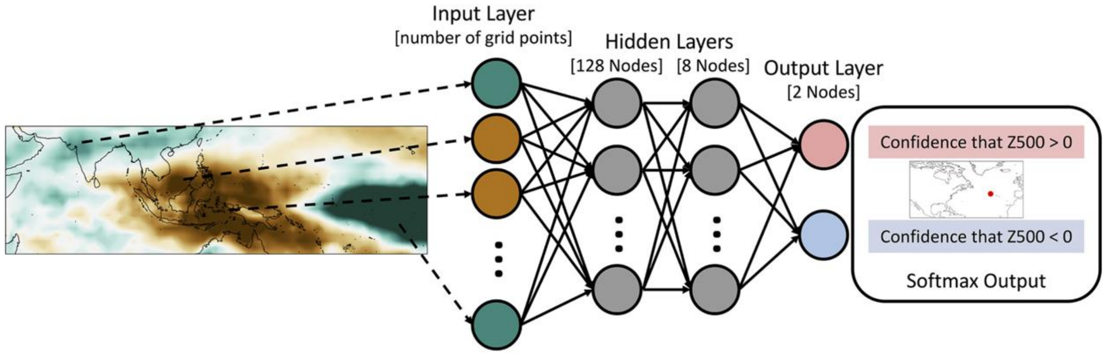
Mamalakis, Antonios, William E Chapman, Kirsten J Mayer. Expanding the Limits of Explainable AI Tools in Generating Physical Insights. *TechRxiv.* November 03, 2025. **DOI: 10.36227/techrxiv.176221224.44249956/v1**

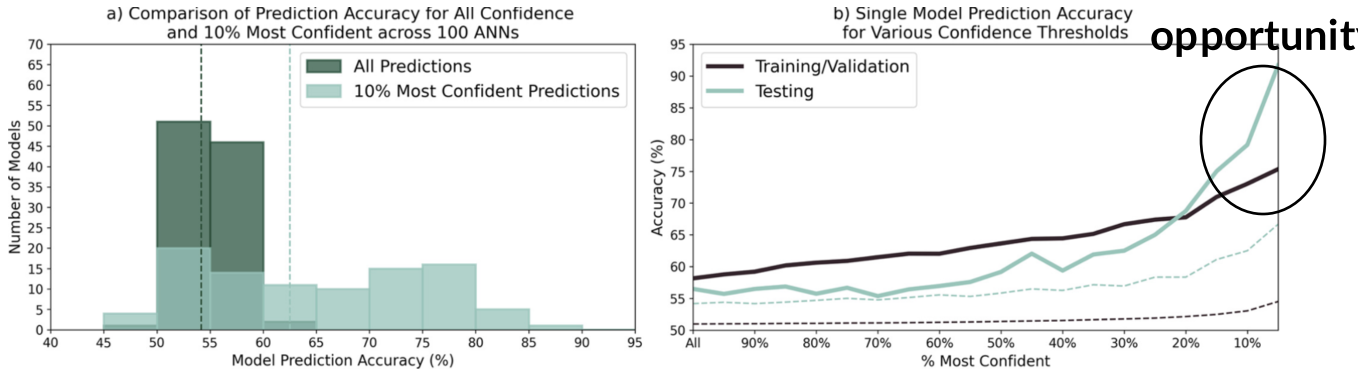
Toms, B. A., Barnes, E. A., & Hurrell, J. W. (2021). Assessing decadal predictability in an Earth-system model using explainable neural networks. *Geophysical Research Letters*, 48, e2021GL093842. **https://doi.org/10.1029/2021GL093842**

Extra

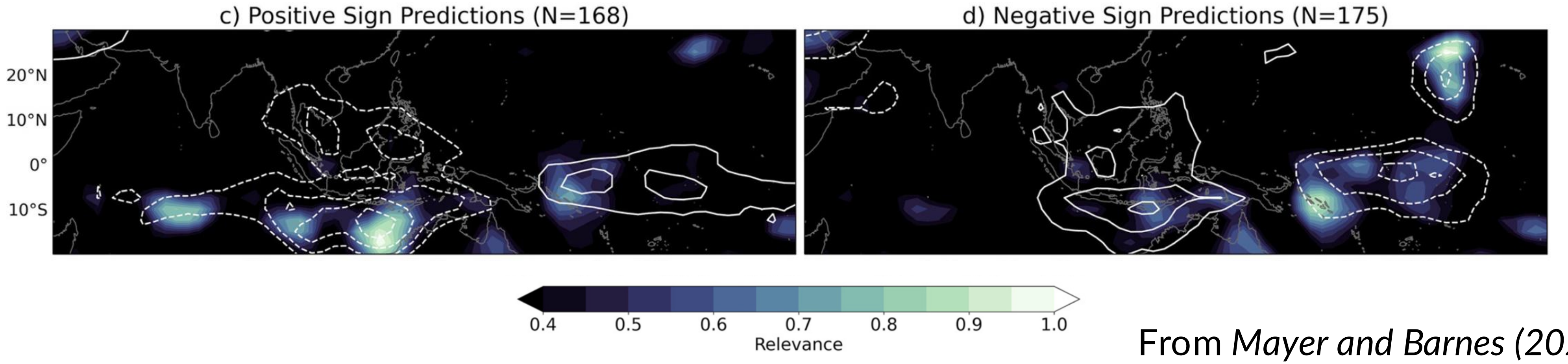
Sub-Seasonal Forecasts of Opportunity

Daily anomalies of Outgoing Longwave Radiation in the tropics are used to predict the sign of Z500 anomaly over North Atlantic after 3 weeks.





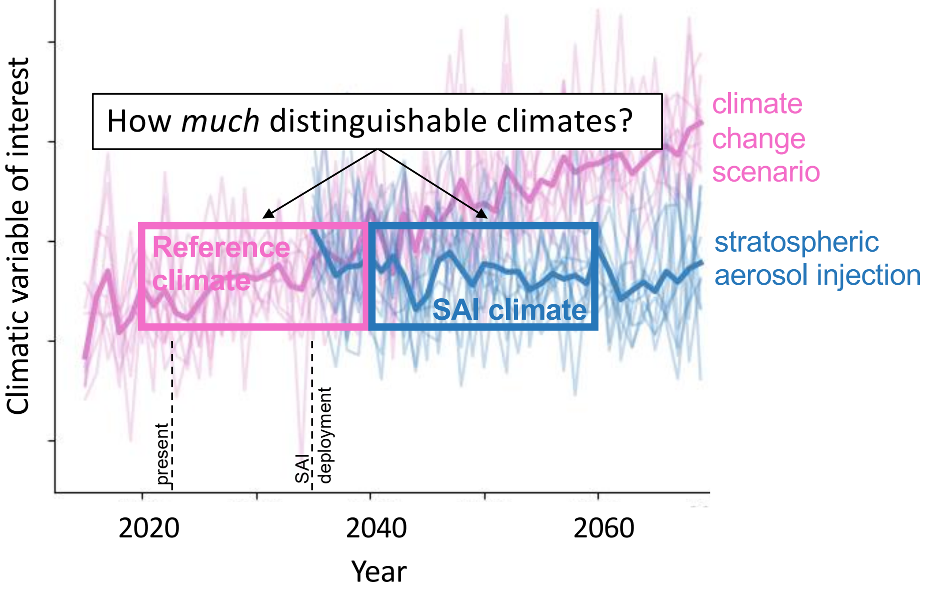
XAI results:



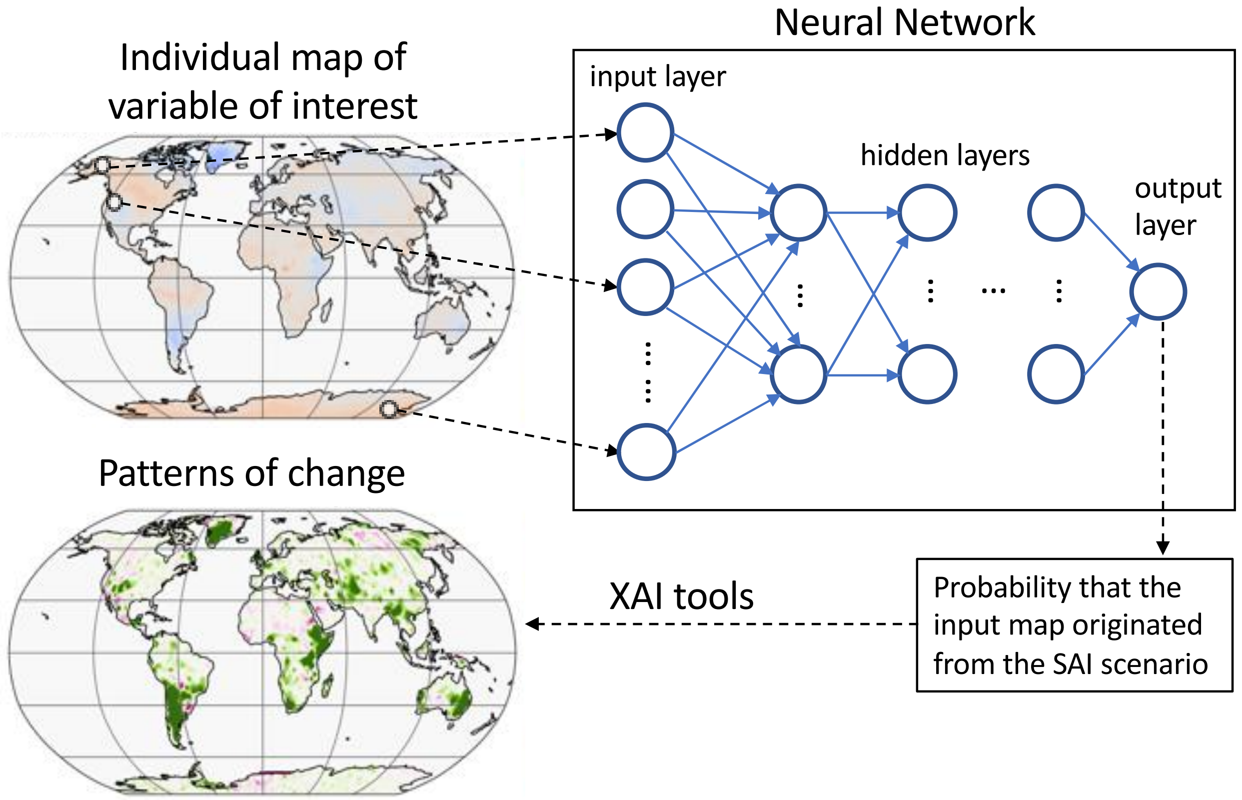
From *Mayer and Barnes (2021)*

Explore indicator patterns of change under Geoengineering Scenarios

Annual maps of precipitation are used to predict whether the map originated from the pre-deployment or the post-deployment climate



XAI results:

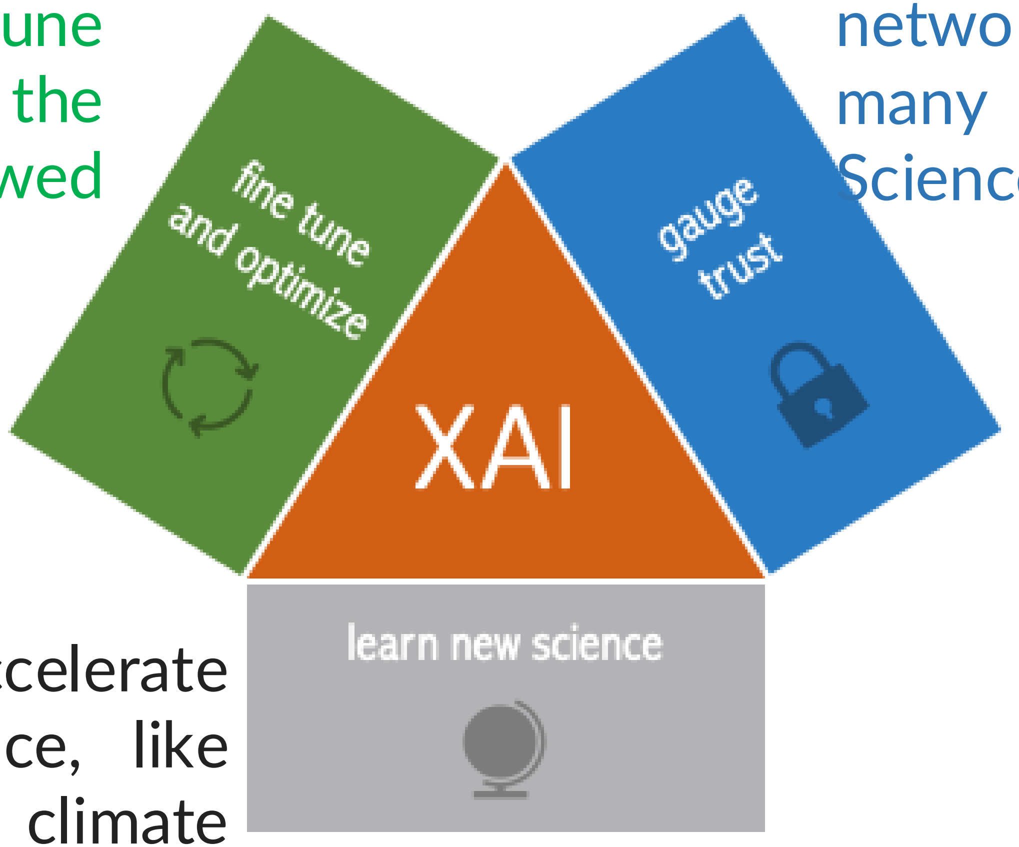


From *Mamalakis et al.* (2023)

XAI: A potential *game changer* for Earth Sciences

XAI helps calibrate model trust and physically interpret the network, which is a necessity in many applications in Earth Sciences.

XAI may help fine-tune and optimize the architecture of a flawed model



XAI may help accelerate establishing new science, like investigating new climate teleconnections and gaining new insights.

From *Mamalakis et al. (2022c)*

Attribution benchmarks for XAI

$F(\mathbf{X})$ is an additive separable function: the summation of local nonlinear responses across the globe:

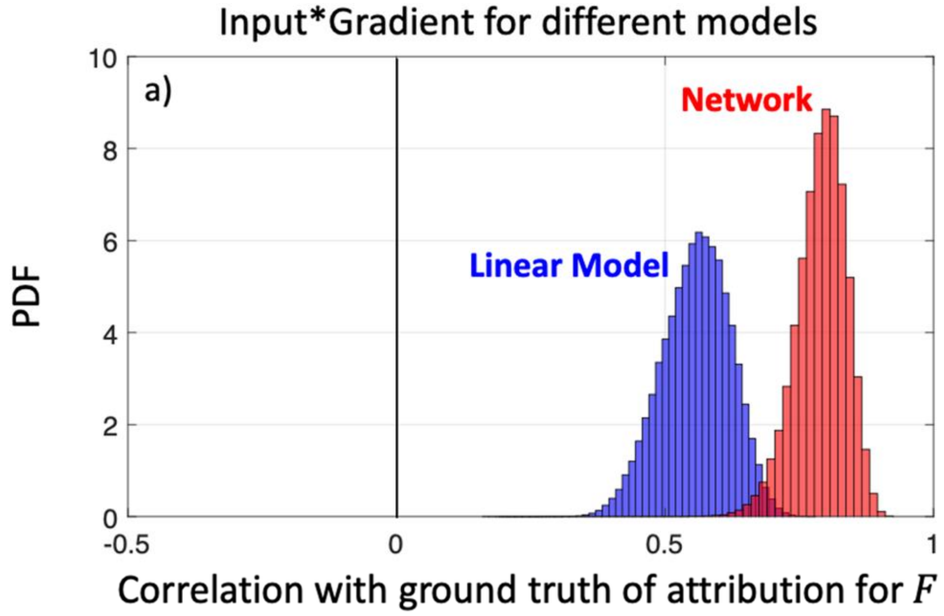
$$Y = F(\mathbf{X}) = F(X_1, X_2, \dots, X_d) = C_1(X_1) + C_2(X_2) + \cdots + C_d(X_d)$$

where, X_i is the random variable at grid point i. Under this setting, the response Y is the sum of local responses C_i across all grid points.

Under this setting, the contribution of variable X_i is by definition equal to C_i.

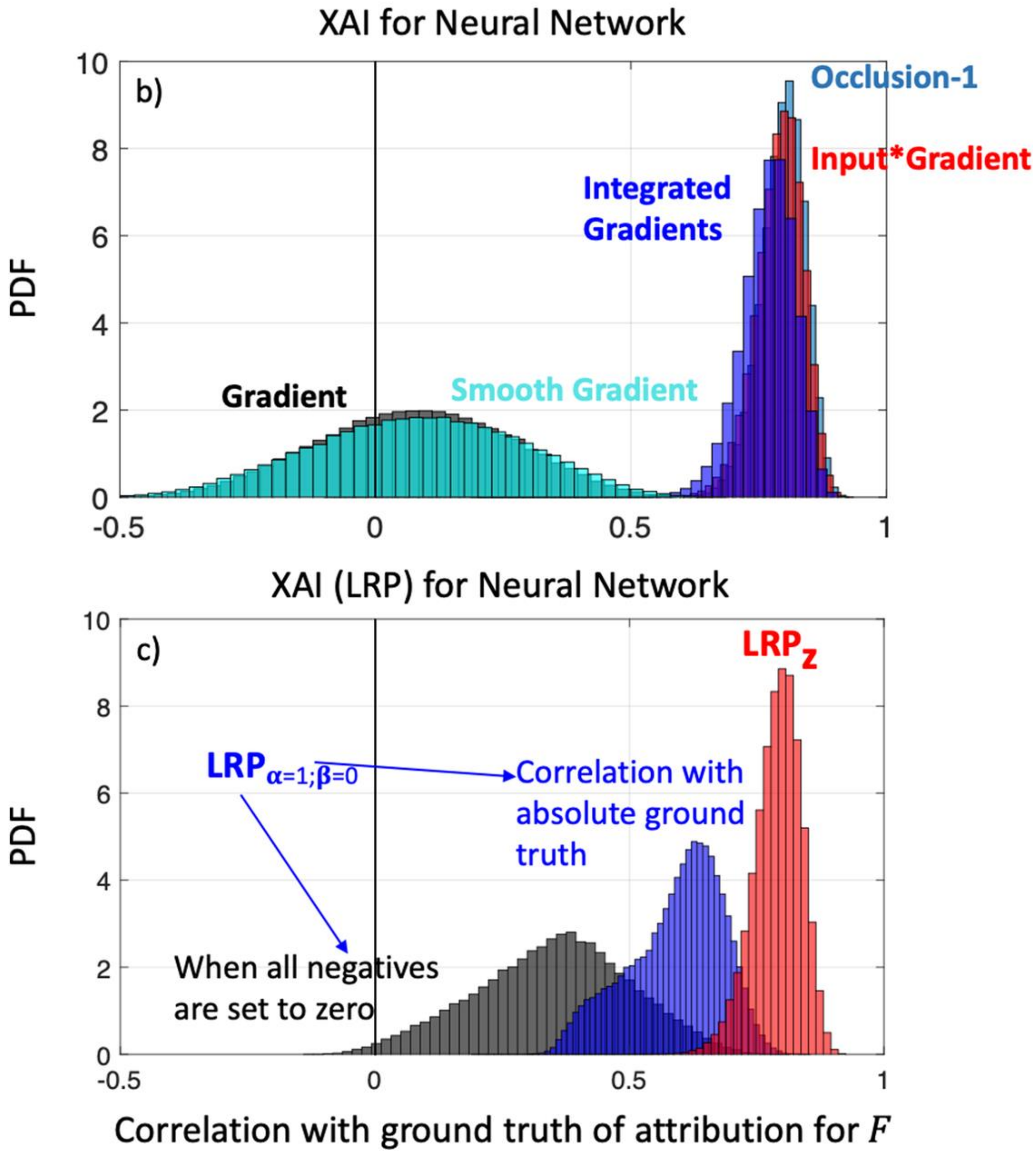
From *Mamalakis et al. (2022)*

Regression Benchmark - Fully Connected Network



- **XAI methods show potential to be a game-changer in how we predict/detect patterns in Earth Sciences. We can use these tools to calibrate model trust, fine-tune models and learn new science.**

- **Given the plethora of methods out there, the lack of a ground truth to assess their fidelity has the risk of allowing subjective assessment, and cherry-picking certain methods. It is important to introduce *objectivity* in XAI assessment and shed light to relative strengths and weaknesses.**



From *Mamalakis et al. (2022)*

Best practices of XAI

Our investigation revealed that likely no universally-optimal method exists. Different methods might be more suitable for different prediction settings and/or network architectures.
We also detected some aspects that need to be considered when applying XAI methods:

i) Gradient shattering is the phenomenon of noisy patterns in the gradient, the level of which is a function of the depth of the network. Gradient shattering might lead to overwhelmingly noisy patterns that make the explanation of any gradient-based method incomprehensible.

ii) Unable to disentangle positive and negative contributions. This may lead to a very distorted picture of what the network's strategy is and possibly limit trust in the predictive model itself.

iii) Ignorant to zero input: Some methods automatically assign a zero attribution to zero values in the input, despite the fact that in specific settings a zero value could be important for the prediction.
* These methods can be perceived as providing attributions by always considering a zero-valued baseline.

From *Mamalakis et al. (2022b)*

Best practices of XAI

Table 1. Summary of XAI methods considered in this study. Practical strengths (✓) and weaknesses (✗) of each method are also reported.

XAI method	Brief summary of the method	Desired property for CNN applications as explored in this study: disentangles the sign of relevance	insensitive to gradient shattering	not ignorant to zero input	Extra comments/insights
Gradient (Simonyan et al., 2014)	Calculates the first partial derivative of the model output with respect to the input. (sensitivity)	✓	✗	✓	Estimates the sensitivity of the output to the input, which is not the same as the attribution; see Appendix B
Smooth Gradient (Smilkov et al., 2017)	Calculates the average gradient across many perturbed inputs. (sensitivity)	✓	✗	✓	
Input*Gradient (Shrikumar et al., 2017)	Multiplies the input with the gradient. (attribution)	✓	✗	✗	
Integrated Gradients (Sundararajan et al., 2017)	Multiplies the average gradient along the straight line between the input point and a reference point with the corresponding distance between the two points. (attribution)	✓	✗	✓	
α1β0	Layer-wise back propagation of each neuron's		✓	✗	Considers only positive preactivations
			✗	✗	Equivalent to Input*Gradient for networks using ReLU activations
			✓	✗	Combines the strengths of LRPz and LRP$_{\alpha1\beta0}$
			✓	✓	Provides a coarser picture of attribution; not suitable if
			✗	✓	based on well-founded theory; computationally expensive

A novel approach to generate datasets with XAI ground truth to evaluate image models

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[b] Laboratory for Artificial Intelligence App Balearic Islands, Dpt. of Mathematics and

CLEVR-XAI: A benchmark dataset for the ground truth evaluation of neural network explanations

Leila Arras[1], Ahmed Osman[1], Wojciech Samek[*]

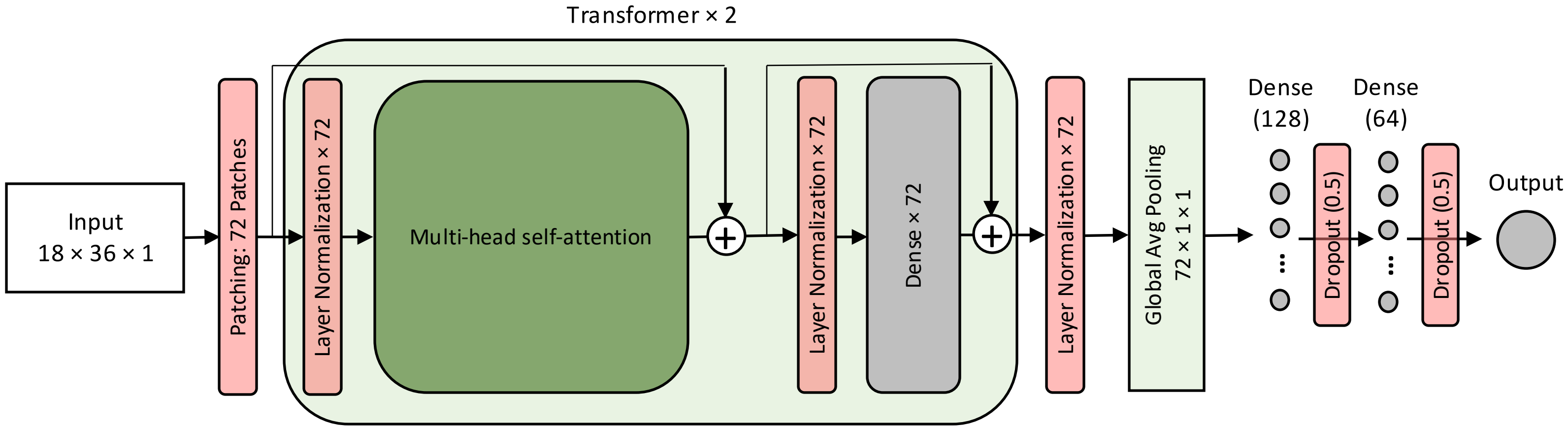
Department of Artificial Intelligence, Fraunhofer Heinrich Hertz Institute, 10587 Berlin, Germany

From *Mamalakis et al. (2022b)*

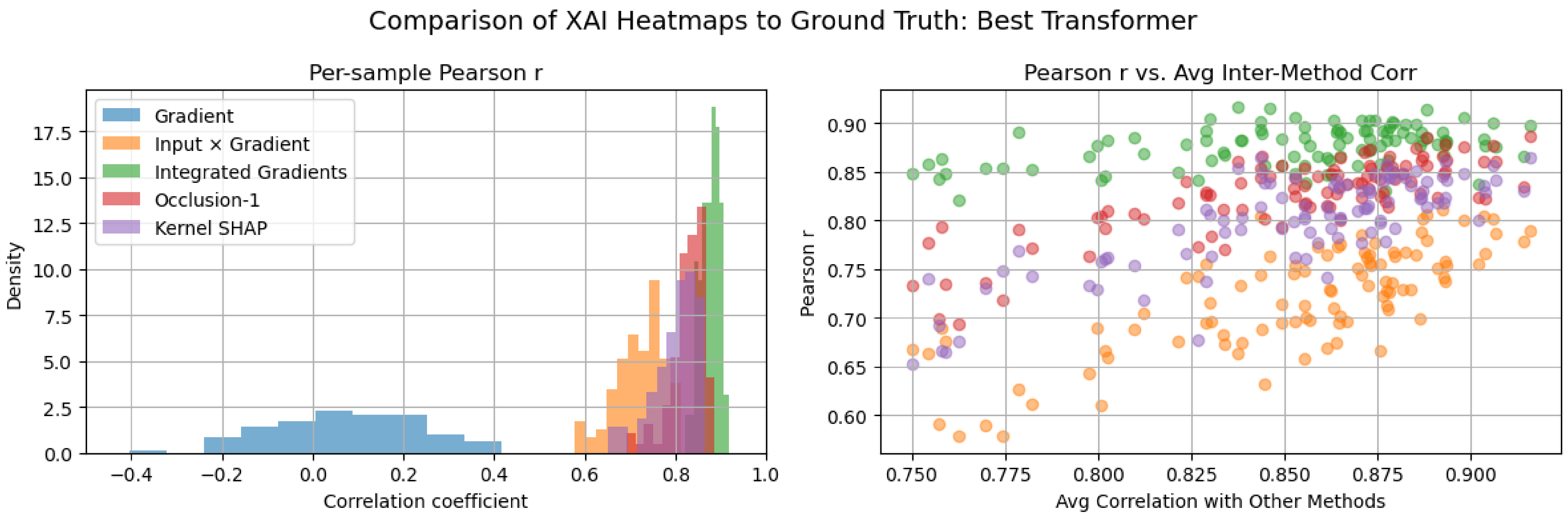
Experiments

From *Mamalakis et al.* (2025)

Transformer Architecture



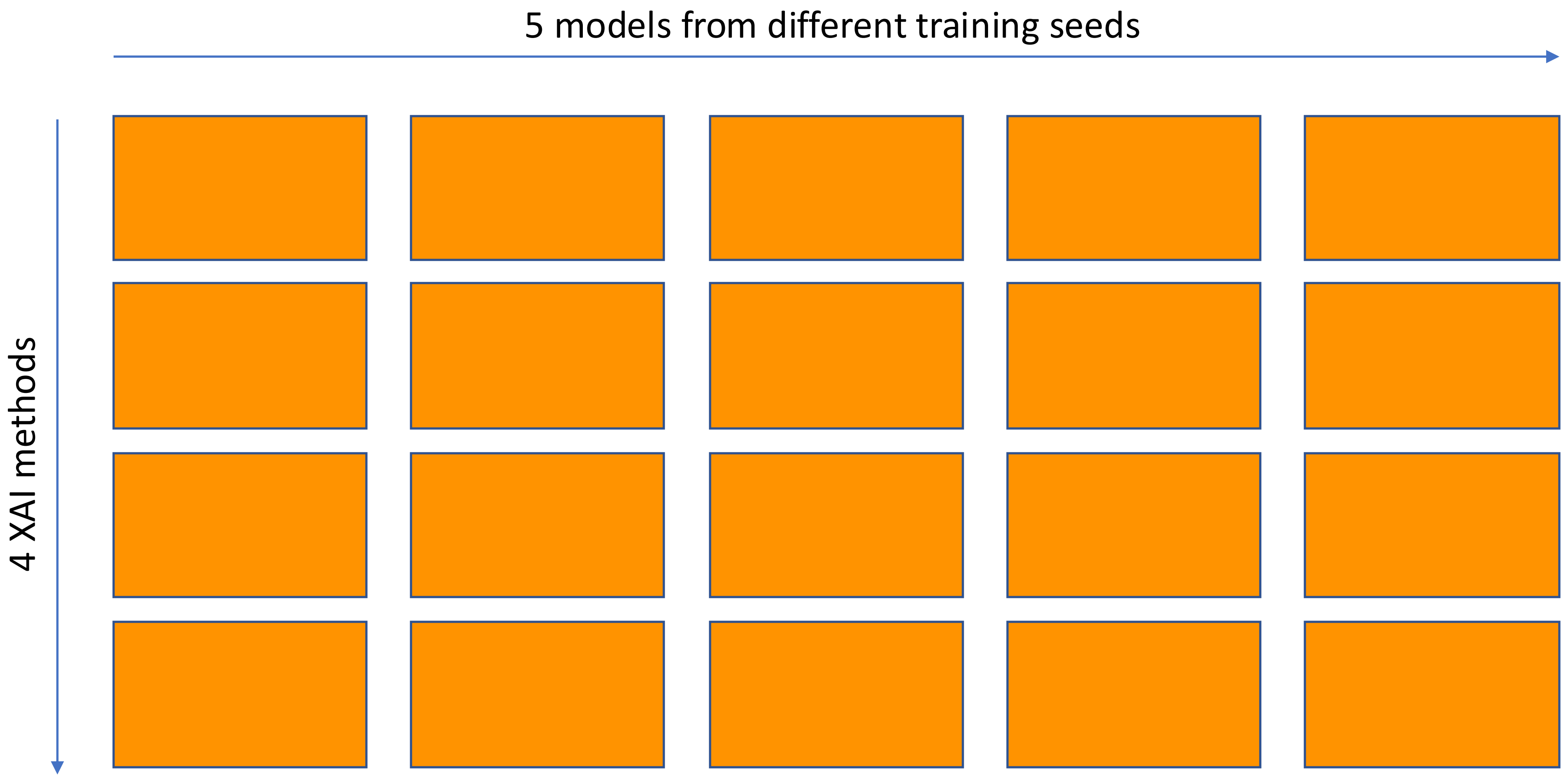
Explaining the agreement with ground truth



Inter-method agreement can be used as a proxy to
indicate attained knowledge across different samples!

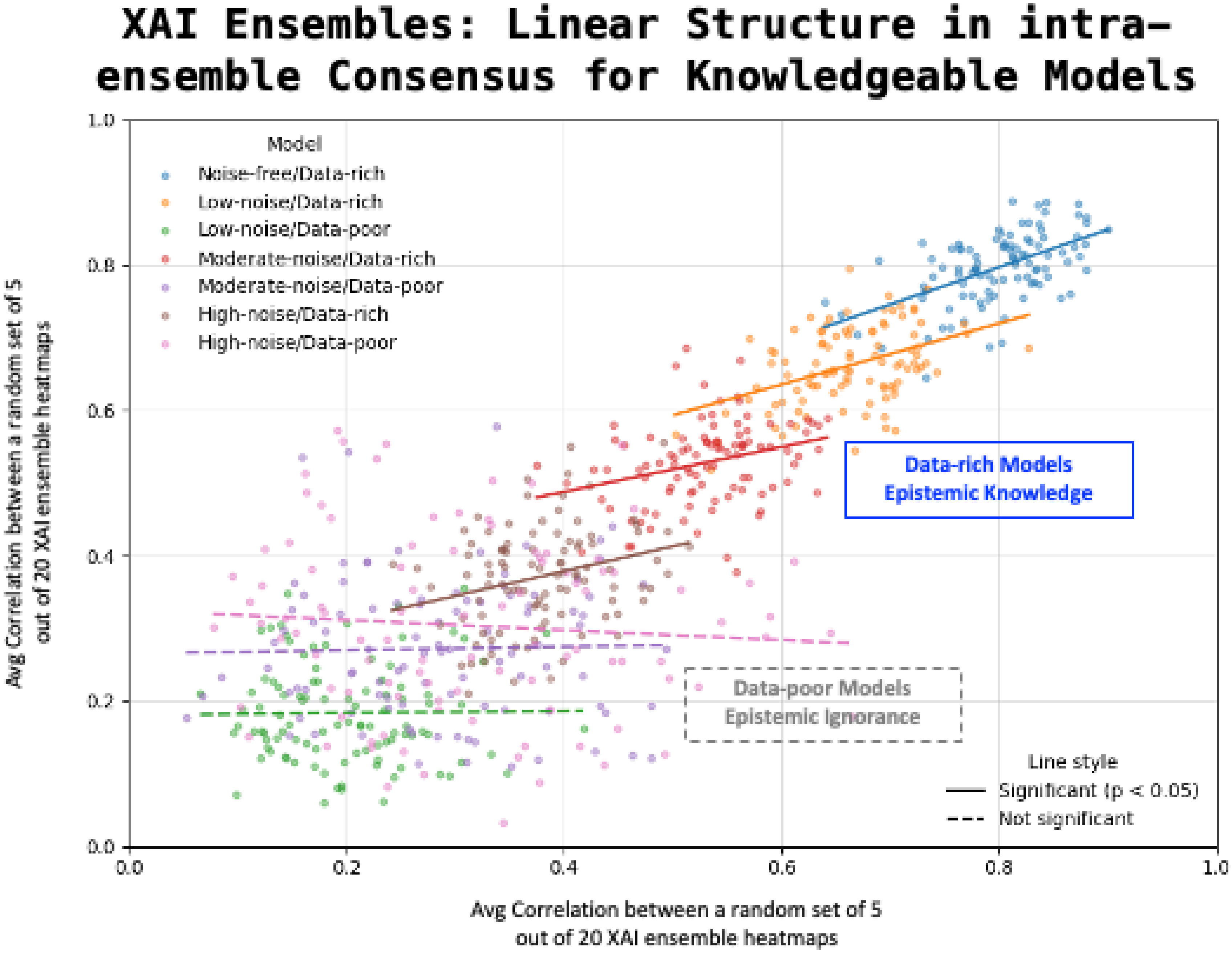
Useful Proxies

Instead of considering inter-method and inter-model variability separately, we consider all XAI heatmaps at once, i.e., the entire **XAI ensemble**:



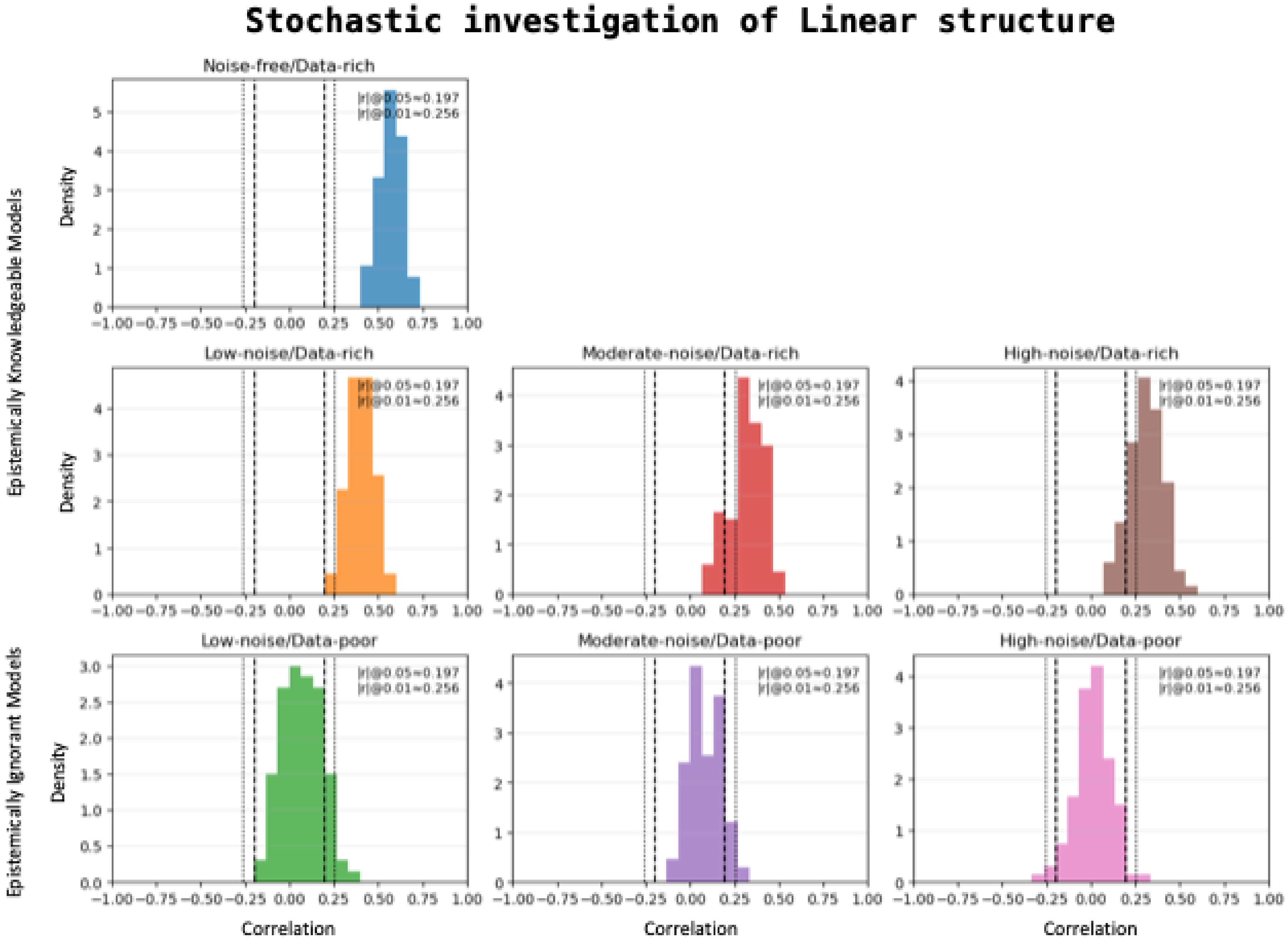
From *Mamalakis et al. (2025)*

Useful Proxies



From *Mamalakis et al. (2025)*

Useful Proxies



From *Mamalakis et al. (2025)*